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Sign Language Recognition by using Spatio-Temporal Features

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Abstract

The recognition of sign language has significant importance as an application that addresses the communication challenges faced by those within the Deaf and hard-of-hearing populations. This study introduces a methodology for the identification of sign language using spatio-temporal attributes, specifically concentrating on two distinct sign language datasets: the CSL (Chinese Sign Language) alphabet and the ArSL (Arabic Sign Language) alphabet. The study's objective is to create a powerful recognition system capable of reliably identifying sign language motions. A combination of spatio-temporal characteristics collected from video sequences of sign language motions is used in the suggested technique. These characteristics capture the spatial layout as well as the temporal dynamics of the motions, allowing the model to recognise signals with more precision. Experiments were carried out on the CSL and ArSL alphabet datasets to validate the method's efficacy. The findings of the trials are encouraging. The suggested system recognized CSL alphabet signs with an accuracy of 90.87% and ArSL alphabet signs with an accuracy of 89.46%. These high accuracy rates show the power of the spatiotemporal feature-based method for sign language identification. The system's success on two different sign languages implies that it is adaptable and useful in a wide variety of sign language applications. This study contributes to the evolution of assistive technology, making sign language recognition more accessible and efficient for those who communicate using sign language. Furthermore, the findings pave the way for additional research into spatiotemporal feature-based approaches in sign language recognition, with potential applications in real-world settings such as sign language interpretation and communication support.

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Keywords: Sign Language;Classification;Spatio-temporal features; Time Distributed CNN;

1. Introduction

Sign language is a crucial means of communication for Deaf and Hard of Hearing people, acting as a bridge to promote connection in a society dominated by spoken language. The capacity of the Deaf population to communicate

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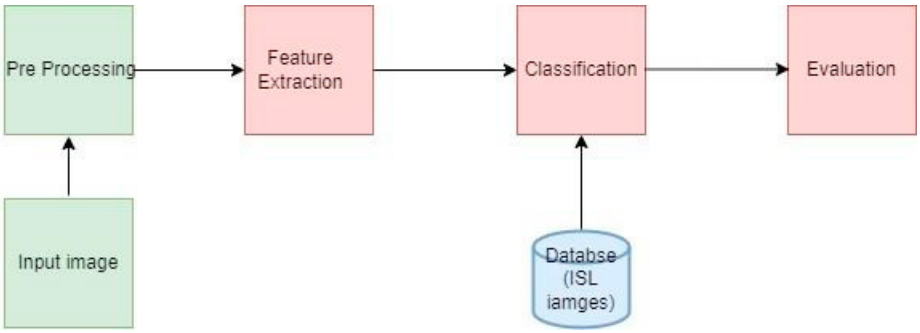


Fig. 1. General Block Diagram of SLRS

successfully using sign language, on the other hand, is dependent on the presence of an interpreter or someone who knows sign language. As a result, the development of automatic sign language recognition systems has become critical in enabling autonomy and inclusion to the Deaf and Hard of Hearing communities.[10],[2] This research goes into the field of sign language recognition, concentrating on an innovative technique that uses spatio-temporal aspects to increase accuracy. Recognising sign language motions provides unique issues since it requires collecting not only static hand forms (spatial characteristics), but also dynamic hand movements.[5],[6] The suggested technique attempts to improve the precision and resilience of sign language recognition systems by incorporating both geographical and temporal information. Experiments on two unique sign language datasets are used to assess the method's efficacy: the Chinese Sign Language (CSL) alphabet and the Arabic Sign Language (ArSL) alphabet. Because these datasets reflect two distinct linguistic and cultural settings, the study is more inclusive and applicable to a wide range of sign languages. This work has considerable significance as it possesses the capacity to revolutionize the recognition and comprehension of sign language. The efficacy of the strategy in enhancing the accessibility and effectiveness of sign language recognition is shown by the notable accuracy rates achieved, namely 90.87% for CSL alphabet and 89.46% for ArSL alphabet. The efficacy of this technique is not limited to particular sign languages, but rather may be used to other sign language systems, thereby making it a flexible and noteworthy addition to assistive technology. Moreover, this study provides opportunities for practical implementation in several domains, including sign language interpretation, communication assistance, and enhanced accessibility for those who are Deaf or Hard of Hearing across diverse contexts. According to [3] , there is a need for a deeper comprehension of The integration of spatiotemporal components facilitates a more comprehensive understanding of the gestures in sign language, hence enhancing the precision of identification and enabling the potential for more seamless and effective communication between those who are Deaf and those who are hearing.

Major contribution of the paper as follows :

- This research presents a novel methodology for sign language recognition that places particular focus on the integration of spatiotemporal information. The integration of spatial and temporal components in technology enhances the accuracy and resilience of sign language recognition, hence facilitating more effective and proficient communication.
- Time distributed CNN model: The CNN architecture used in the model is time distributed. This architectural design enables enhanced identification capabilities by incorporating the temporal dynamics of ISL motions via the analysis of sequential video frames.
- Cross-Linguistic Applicability: This study illustrates the method's cross-linguistic applicability by testing it on two unique sign language datasets, CSL and ArSL. The method's effectiveness in recognising multiple sign languages shows that it is adaptable to a wide range of linguistic and cultural situations, potentially increasing its worldwide influence.
- High Accuracy Rates: The study's accuracy rates for CSL and ArSL alphabet sign recognition were 90.87% and 89.46%, respectively. These high accuracy levels demonstrate the method's ability to reliably detect sign language motions, making it a vital contribution to the field of assistive technology.

Fig 1 shows the General block diagram of SLRS. The following explains the steps involved in the SLRS

- Input image : The procedure starts with an acquired image or a series of images depicting sign language motions.
- Preprocessing: It encompasses a series of activities aimed at improving the quality and appropriateness of input pictures for subsequent analysis. Common Preprocessing Steps are:
  - Resizing: To maintain visual uniformity, it is important to ensure that all images are resized to a constant size.
  - Normalization :Normalization refers to the process of modifying pixel values in order to conform to a standardized scale.
  - Noise reduction:It involves the removal or reduction of artifacts or noise present in a picture.
  - Color Conversion: It is in cases when color is not crucial, photographs may be converted to grayscale.
- Feature Extraction: The process of feature extraction involves the identification and capture of pertinent information or patterns from preprocessed pictures. Common Techniques are
  - Convolutional Neural Networks(CNN): CNN has the ability to autonomously acquire hierarchical features.
  - The Histogram of Oriented Gradients (HOG): HOG technique is used to depict the gradients within certain parts of a picture.
  - The Local Binary Patterns (LBP): LBP method is used to characterize and describe local patterns within a given texture.
- Classification : Classification is a fundamental job in which a system provides a label or class to an input picture, based on the extracted characteristics. Commonly used classification models are.
  - Support Vector Machines: SVM is a popular machine learning algorithm widely used in many fields. SVMs are a kind of supervised learning model that can be used for both classification and regression tasks.
  - The K-Nearest Neighbors: KNN algorithm assigns a label to a data point by determining the majority classamong its nearest neighbors.
  - Deep learning models: DL models such as Convolutional Neural Networks (CNNs), are capable of acquiring hierarchical representations to facilitate the task of categorization.
- Evaluation: The evaluation process involves the assessment of the sign language recognition system's performance via the use of metrics that measure its accuracy and effectiveness. Commonly used evaluation metrics are
  - Accuracy: It refers to the measure of accurately categorized cases in relation to the total number of examples.
  - Precision: It refers to the degree of precision shown by favorable forecasts.
  - Recall : The concept of recall, also known as sensitivity, refers to the capacity to accurately identify and record all occurrences that are positive.
  - F1 Score: The F1 Score is a metric that provides a balanced evaluation of accuracy and recall.

The subsequent section provides a comprehensive analysis of prior scholarly contributions in the field, before the subsequent discussion on the chosen research approach. Subsequently, the Results section ensues, and subsequently, the piece is brought to a conclusion.

2. Related Works

[7] provides a comprehensive evaluation of the current body of work on automated sign language processing. The assessment covers several aspects including the devices used for sign language learning, the databases available for research purposes, and the methodologies used for recognizing different types of signs such as fingerspelling signs, isolated sign words, and continuous sentence recognition systems.

They addresses current breakthroughs in sign language processing, such as deep machine learning and multimodal techniques, as well as various related difficulties. [1] Provides a graph-matching model with an early estimation technique for addressing the issues of detecting signs with varying amounts of motion frames and extracting signs from non-sign hand movements in 3D sign language recognition. The model undergoes verification using benchmark datasets and state-of-the-art graph matching methods. It is then tested for its accuracy and efficiency in the identification of 3D signs, taking into consideration real-time limits. The evaluation of the Early Estimation Model with Graph Matching (EEMGM) approach is conducted by analyzing precision-recall curves, percentage recognition rate, and the computational time required for sign recognition. The precision, recall, and recognition rate measures are used to assess the model's accuracy and efficiency. Four tests, involving variations in frame rates, sign movements, and cluttered hand gestures, are aimed to test the efficiency and durability of the proposed system. The validity of the proposed model is assessed by using a dataset consisting of 350 Indian sign language phrases, as well as five benchmark datasets. The findings indicate that the use of the Early Estimation Model using graph matching techniques improves the efficiency and precision of 3D sign language identi- fication. Moreover, this approach effectively tackles challenges related to diverse frame rates and the presence of hand clutter. [11] conducted a study on recognition. This work introduces an innovative methodology for sign language detection via the use of computer vision techniques and deep learning algorithms. The Sign Language Recognition System (SLRS) is a proposed system designed to improve communication between those who possess sign language proficiency and those who do not. This study focuses on three main contributions: the creation of a comprehensive dataset of Indian sign language (ISL) consisting of 65 different users, the expansion of the dataset to increase variabil- ity within the same class, and the proposal of a robust model that utilizes Convolutional Neural Network (CNN) for feature extraction and classification of ISL gestures. The evaluation of the method's efficiency is conducted by con- sidering precision, recall, f-score, and the time required by the system. The results indicate promising performance when compared to previous research efforts. [8] SignQuiz is an economically efficient online program designed for the acquisition of fingerspelled signals in Indian Sign Language (ISL) via the use of automated sign language recognition (ASLR) methodology. The use of this technology obviates the need for external sensors and offers a superior educa- tional encounter in comparison to traditional printed materials. This research examines the progression of SignQuiz, the use of deep neural networks for acquiring fingerspelled signs, and the favorable outcomes that demonstrate its efficacy when compared to conventional approaches. The SignQuiz application demonstrated high recognition accuracies when using the InceptionV3 and Nasnet models. . The evaluation of the literature in this section leads to the following conclusions:

- Limited and diverse datasets: A prevalent obstacle in the field of SL gestures is the restricted availability of datasets that are both limited in size and diversified in kind. Numerous scholarly articles discuss the process of dataset production; yet, there is room for improvement in terms of the scale, variety, and quality of the datasets.
- Pre-processing techniques: Pre-processing procedures are crucial in enhancing the quality of input data. Further investigation and advancement in the realm of advanced preprocessing techniques tailored specifically for the identification of gestures is necessary.
- Feature extraction: The ongoing research problems in the field of feature extraction and representation are to the identification and extraction of relevant variables that effectively capture the unique characteristics of sign language motions.

3. Proposed Work

The research employs a specialized Time Distributed Convolutional Neural Network (TD-CNN) structure specifically tailored for analyzing the Sign-to-Text System. The focus of this study is on two publicly available datasets released by Alphabet. There are a total of 32 unique sets of alphabets inside each dataset. The following parts provide an exhaustive examination of the deep learning approach used in our study, along by a synopsis of the datasets employed. Fig 3 illustrates the overarching structure of the Convolutional Neural Network (CNN) model. The conversion of an existing Convolutional Neural Network (CNN) model into a Time Distributed CNN model, with the ability to handle 32 unique classes, requires changes to both the architecture and output layer. These modifications are implemented to consider the time aspect and the exact quantity of classes associated with the activity. This section provides a description of the process by which the current Convolutional Neural Network (CNN) architecture is adapted and transformed into a Temporal-Difference Convolutional Neural Network (TD-CNN).

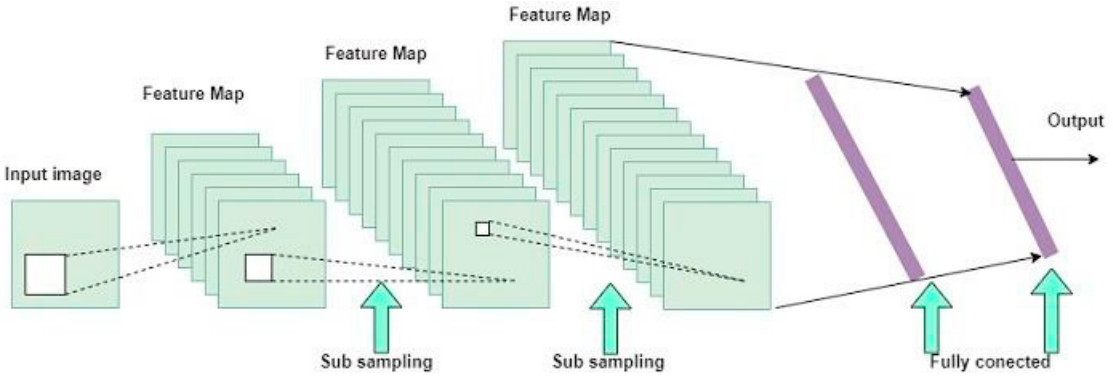


Fig. 2. General Architecture of CNN

Algorithm 1 Time Distributed CNN

Input: Sequence of frames (X) with shape (num_frames = 5, $height$ = 64, $width$ = 64, $channels$ = 1)
Output: Predicted labels
procedure DefineTimeDistributedCNN
    Create a sequential model.
    Add a Time Distributed 3D Convolutional layer with appropriate parameters:
        - Convolutional filters=64, kernel size=3 X 3, activation function=relu.
    Add a Time Distributed 3D MaxPooling layer.
    Flatten the output before fully connected layers.
    Add fully connected layers with appropriate activation functions.
    Compile the model with loss function = crossentropy, optimizer=Adam, and metrics=Accuarcy.
end procedure
procedure TrainTimeDistributedCNN
    Split the dataset into training and testing sets.
    Preprocess the input data (normalization, resizing).
    Train the model using the training set:
        Use `model.fit()` with the training data and labels.
    Evaluate the model on the testing set:
        Use `model.evaluate()` with the testing data and labels.
end procedure
procedure MakePredictions
    Preprocess new input data if needed.
    Use the trained model to make predictions on new data:
        Use `model.predict()`.
end procedure
Output the final predictions.

3.1. Proposed Model

The adapted model has been specifically developed to cater to sequence-to-sequence tasks, whereby the input data comprises a series of two-dimensional images, and the desired output is a sequence of corresponding labels. Algorithm 1 represents the steps involved in the proposed model. The model is composed of the following layers.

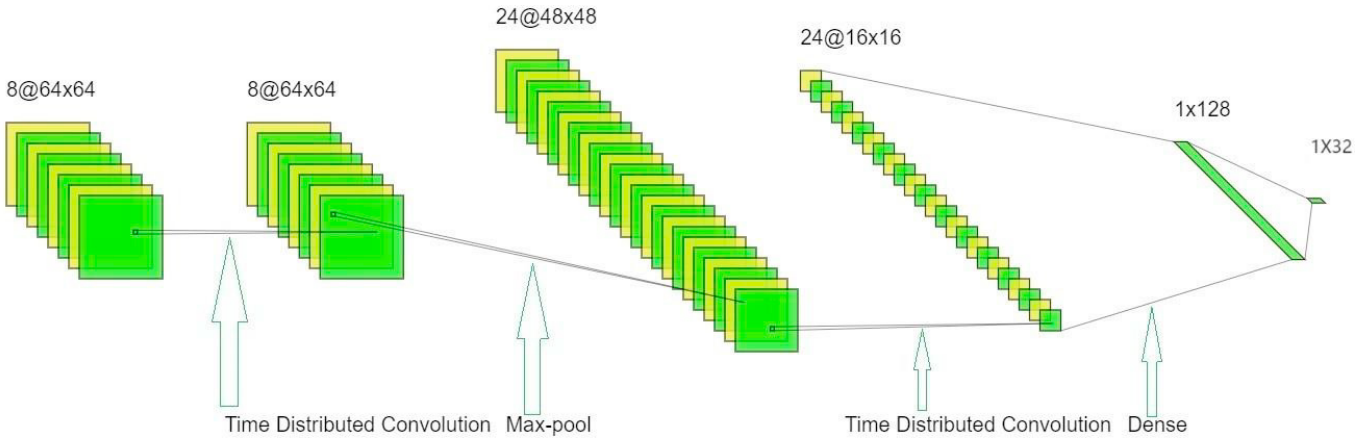


Fig. 3. Time Distributed CNN Model

- Time-Distributed Convolutional Layer: The present layer provides a two-dimensional convolution operation on every time step of the input sequence, using a kernel size of 3x3 and a total of 64 filters. The Rectified Linear Unit (ReLU) activation function is used in order to include non-linearity.
- Time-Distributed Max Pooling Layer: The time-distributed max pooling layer is a commonly used element within deep learning architectures designed for the study of temporal data. The TimeDistributed layer is utilized in conjunction with the MaxPooling2D layer.The aforementioned layer performs 2D max pooling on every time step inside the sequence. The downsampling process reduces the computing strain by decreasing the spatial dimensions of each picture.
- The Time-Distributed Flatten Layer: The Time-Distributed Flatten Layeris a component used in deep learning models that operates on sequential data. It is designed to transform the input data, which is often a sequence. The function of this particular layer is to compress the output of the convolutional layers at each individual time step, hence facilitating the further processing of the data by the LSTM layers.
- LSTM Layer: The first LSTM layer with 64 units, configured to return sequences, is then followed by another LSTM layer with 64 units. LSTM layers are prevalent in the modeling of sequential data. The first LSTM layer is configured with the parameter, which signifies that it generates sequences as outputs rather than a singular output. The sequential output from the first LSTM layer is processed by the second LSTM layer.
- Dense Layer: These layers, sometimes referred to as fully linked layers, are a basic element within deep learning models. The model consists of two dense layers. The first layer is composed of 128 units and utilizes a rectified linear unit (ReLU) activation function. The second layer has a number of units according to the number of classes present in the dataset and employs a softmax activation function. The fully connected layers follow the LSTM layers. The first dense layer is composed of 128 units using a rectified linear unit (ReLU) activation function, whereas the final dense layer consists of 32 units representing the number of classes and applying a softmax activation function. The softmax function is often used in classification tasks to provide a probability distribution over many classes.

  The reaction of the proposed model to a big dataset is generally positive, leading to improved generalization, learning, and robustness. However, careful consideration of resource requirements and the task at hand is necessary to achieve optimal results. Additionally, monitoring model performance on validation or test sets is crucial to ensure that the benefits of a large dataset are effectively realized.

Fig 3 shows the architecture of proposed Time Distrubuted CNN model



Fig. 4. Images of ArSL Dataset [13]

3.2. Dataset Description

3.2.1. Arabic Sign Language Dataset(ArSL) Alphabet

[13]The dataset known as the RGB Arabic Alphabet Sign Language (ArSL) collection has a total of 7,857 unprocessed images that have been fully annotated with RGB tags. These images depict the alphabets of Arabic sign language and, to the best of our knowledge, represent the first publicly available RGB dataset of its kind. The dataset has been developed with the purpose of aiding persons in the creation of practical Arabic sign language classification models. The data for the American Association of School Librarians (AASL) study was collected from a diverse sample of more than 200 persons, including various situations such as lighting conditions, background settings, picture orientations, image sizes, and image resolutions. In order to ensure the integrity of the dataset, professionals in the field oversaw, validated, and refined the collected images. Fig 4 shows the sample images of ArSL dataset.

3.2.2. Chinese Sign Language Dataset (CSL)

The CSL Alphabet refers to a compilation of visual representations that depict the signs used in the fingerspelled alphabet of Chinese Sign Language (CSL) dataset [14]. Each symbol is associated with a certain Chinese alphabet letter. The collection comprises a sequence of visual representations showcasing individuals proficient in native Chinese Sign Language (CSL) effectively demonstrating the CSL fingerspelling alphabet using sign gestures. The collection has a total of 360 videos, with each letter of the CSL fingerspelling alphabet represented by 30 films. Each video has an approximate duration of 5 seconds and exhibits the signer's execution of the letter from many perspectives. Figure 5 displays a collection of example images from the CSL alphabet dataset.

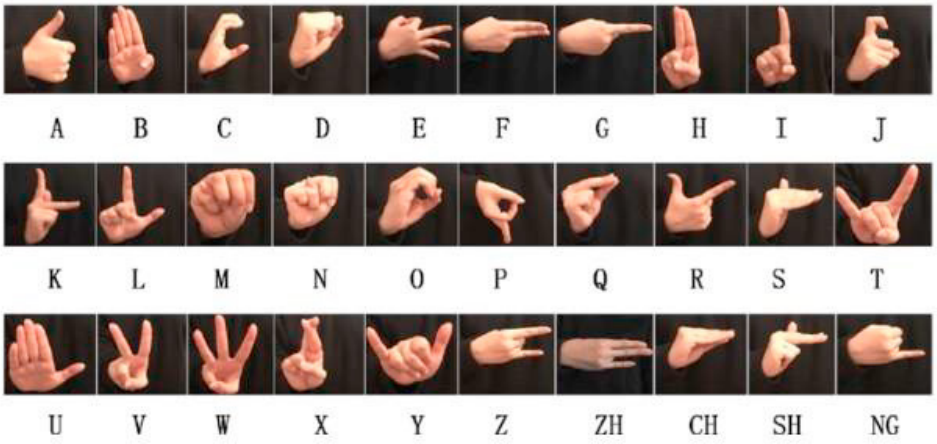


Fig. 5. Images of CSL Dataset [14]

4. Results and Discussion

The experimental setup incorporates the utilization of Google Colab, an AMD Ryzen central processing unit, and Radeon graphics accompanied by a random-access memory (RAM) capacity of 24 GB.

The following section presents an analysis of the outcomes obtained for each dataset.

4.1. Arabic Sign Language (ArSL) Dataset

Table 1 presents a concise summary of the analytical findings acquired using the proposed Time Distributed Convolutional Neural Network (CNN) model. The findings of this study suggest that the model demonstrated a recognition accuracy of about 89.46% while classifying signs from the ArSL dataset.The model has a significant capability in successfully identifying positive situations, particularly in effectively recognizing indicators, as seen by its accuracy score of 88.82%. The recall score of 88.21% indicates the model's effectiveness in correctly recognizing a substantial part of the relevant positive cases in the dataset. The F-score, with a value of 87.48%, functions as a comprehensive statistic that combines accuracy and recall, so providing an assessment of the overall effectiveness of the model. Figure 6 displays the figure illustrating the accuracy and loss metrics of the ArSL Dataset.

Table 1. Performance Evaluation of Proposed Model

Dataset	Accuracy (%)	Precision (%)	Recall (%)	F-Score (%)
ArSL	89.46	88.82	88.21	87.48
CSL	90.87	90.23	89.49	88.64

4.2. Chinese Sign Language Dataset (CSL)

In the context of the CSL dataset, the model demonstrated a significantly improved level of accuracy, reaching a value of 90.87%. The accuracy rate of 90.23% indicates that the model has a high level of proficiency in accurately classifying positive instances, specifically in detecting indicators within the CSL dataset. The recall score of 89.49% indicates that the model effectively identifies a substantial proportion of the relevant positive instances. Nevertheless, it is crucial to acknowledge that the F-score of 88.64% exhibits a modest decrease compared to the other measures, maybe attributable to the equilibrium between accuracy and recall. The findings indicate that the sign language recognition model exhibits satisfactory performance on both datasets, with elevated accuracy as well as robust precision and

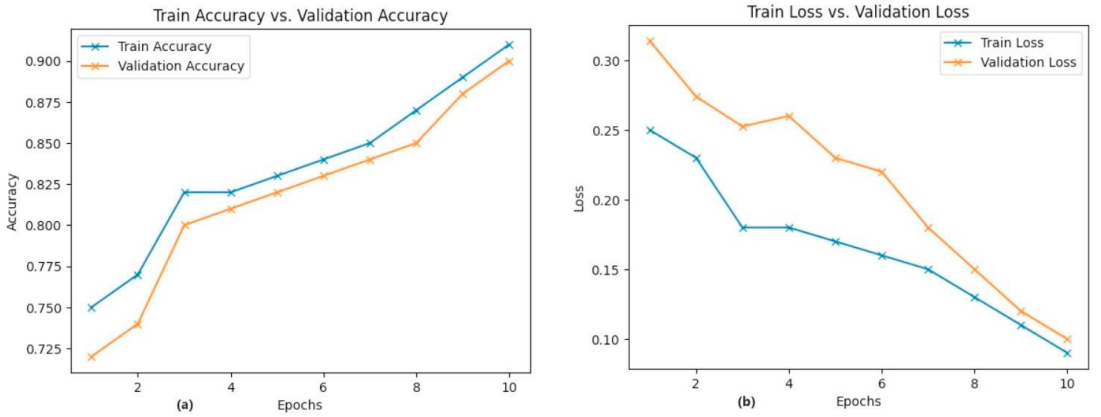


Fig. 6. (a) Accuracy over Epoch plot of ArSL Dataset (b) Loss over Epoch of ArSL Dataset

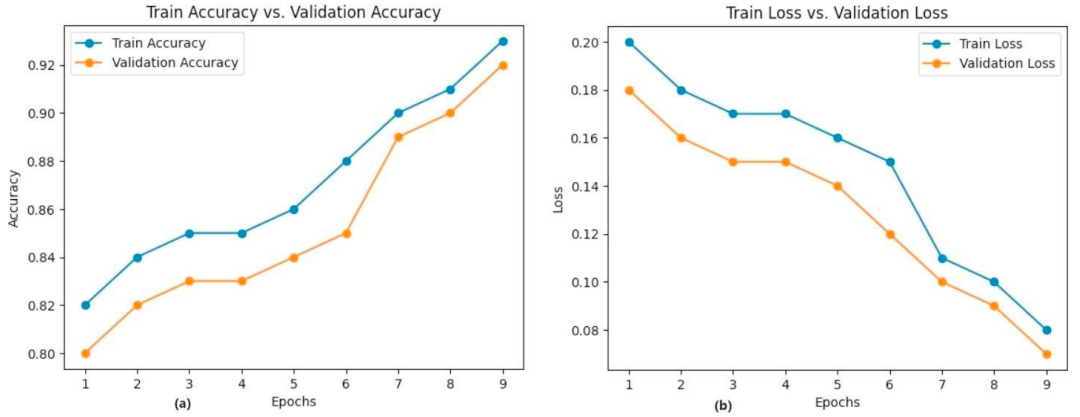


Fig. 7. (a) Accuracy over Epoch plot of CSL Dataset (b) Loss over Epoch of CSL Dataset

recall metrics. When assessing the performance of recognition systems, it is crucial to take into account the unique objectives of the system and the trade-offs between accuracy and recall. Fig 7 shows the accuracy and loss plot of CSL Dataset

4.3. Comparison with state of art models

[9]The applied model utilizes a two-dimensional convolutional neural network (CNN) architecture, which is presumably tailored to effectively capture spatial characteristics included in static pictures of sign language. Attaining an accuracy rate of 88.7% demonstrates a notable proficiency in discerning patterns and characteristics within the ASL Fingerspelling dataset. [15]The 3D convolutional neural network (CNN) model is expected to take into account both spatial and temporal dimensions, making it well-suited for the recognition of dynamic motions in movies of sign language. The observed increase in accuracy of 89.3% relative to the 2D CNN indicates the efficacy of capturing temporal dependencies. [12] The hybrid convolutional neural network (CNN) - long short-term memory (LSTM) model integrates the advantages of both architectures, effectively using spatial and temporal information. The achieved accuracy of 89.1% indicates enhanced performance in comparison to the individual 2D CNN and LSTM models. [4] The utilization of transfer learning with a robust architecture such as DenseNet201 demonstrates that the incorporation of pre-trained models learned on extensive datasets is advantageous for proficient feature extraction. The utility of transfer learning in SLR is shown by an accuracy rate of 88.3%. The findings underscore the efficacy of various structural

designs in the context of speech recognition, emphasizing the significance of taking into account both spatial and temporal elements. The findings underscore the efficacy of various structural designs in the context of speech recognition, emphasizing the significance of taking into account both spatial and temporal elements. The better accuracy of the proposed model highlights the importance of effectively capturing temporal relationships in sign language motions.

5. Conclusion and Future Direction

The use of spatial-temporal elements in the sign language recognition model has shown encouraging outcomes when applied to the ArSL and CSL datasets. The model's proficiency in identifying signals from sign language videos is shown by attaining an accuracy rate of 89.46% for the ArSL dataset and 90.87% for the CSL dataset. The obtained accuracy values of 88.82% and 90.23% for ArSL and CSL, respectively, demonstrate the model's proficiency in effectively categorizing positive situations. This capability holds significant importance within the domain of sign language recognition. The recall scores of 88.21% and 89.49% provide as proof of the model's effectiveness in reliably detecting positive instances that are relevant, hence further substantiating its practical use. The F-scores of 87.48% and 88.64% indicate a balanced assessment of accuracy and recall, highlighting the overall effectiveness of the model. The future holds considerable promise for sign language recognition, since it has the capacity to effectively address communication hurdles and promote inclusivity. Through continuous research and advancement, the previously described model has the capacity to function as a substantial asset for persons belonging to the Deaf and Hard of Hearing population. Moreover, the potential uses of this technology may be seen in other fields, including but not limited to education, communication devices, and accessibility solutions.

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